

Provenance & field-standard check — $M_{\bullet}-\sigma$ / $M_{\bullet}-M_{\star}$ offset significance (2026-06-17)

Backs `scripts/relation_offset_significance.py` → `results/relation_offset_significance.npz` (TESTS O1). Records (1) where the Greene+2020 relation parameters come from and how they were verified, and (2) that the “how-many- σ -from-the-relation” methodology matches field standard.

1. Relation parameters — verified against Greene, Strader & Ho (2020)

Reference: Greene, Strader & Ho 2020, ARA&A 58, 257, “Intermediate-Mass Black Holes” (arXiv:1911.09678; DOI 10.1146/annurev-astro-032620-021835). Zotero key S7XU8UGY.

Functional form (verified verbatim in the paper main text, §8 Scaling Relations): “Assuming that $\log(M_{\text{BH}}/M_{\odot}) = \alpha + \beta \log(\sigma^*/160 \text{ km s}^{-1}) + \epsilon$, where ϵ is the intrinsic scatter, we present our fits in Supplemental Table 5 and Figure 3.”

and the $M_{\bullet}-M_{\star}$ caption (Supplemental Table 5): “ $\alpha + \beta \log(M^*/M_0) + \epsilon$ for $M_0 = 3 \times 10^{10} M_{\odot}$ ”.

Adopted coefficients = Greene+2020 Supplemental Table 5, “Early” (early-type) subsample — the correct sample for AGEL J0206’s deflector (massive, passive, E/S0). Confirmed against BOTH the in-repo machine table `../AGEL_Mbh_sigma/Greene20_Supple_5.csv` AND the supplemental PDF `../AGEL_Mbh_sigma/aa58_greene_supmat.pdf` (Supplemental Table 5, full uncertainty columns):

Fit	Sample	α	β	ϵ (intrinsic scatter)	pivot
$M_{\bullet}-\sigma^*$	Early	8.03 ± 0.06	4.24 ± 0.25	0.43 ± 0.04	$\sigma/160 \text{ km s}^{-1}$
$M_{\bullet}-M^*$	Early	7.89 ± 0.09	1.33 ± 0.12	0.65 ± 0.05	$M^*/3 \times 10^{10} M_{\odot}$

(For context, the other Supplemental-Table-5 rows: $M_{\bullet}-\sigma$ All-no-limits 7.88/4.34/0.53, All-limits 7.87/4.55/0.55, Late-no-limits 7.40/2.54/0.50, Late-limits 7.44/3.61/0.58; $M_{\bullet}-M^*$ All-no-limits 7.56/1.39/0.79, etc. We use Early throughout, matching the E+S0 comparison sample plotted in the paper figures and the deflector’s morphology.)

These are exactly the values hard-coded in the paper figure notebook (`AGEL_0206_ApJL_Figures/figures_paper4_2026-06-08.ipynb`, $M_{\bullet}-\sigma$ / $M_{\bullet}-M_{\star}$ cells) and in `scripts/relation_offset_significance.py` — so the significance calc, the figures, and Greene+2020 are mutually consistent. [?](#)

Verification trail (2026-06-17): Zotero fulltext of S7XU8UGY confirmed the $M_{\bullet}-\sigma$ functional form + $\sigma/160$ pivot; web cross-check confirmed the $M_{\bullet}-M_{\star}$ Early row (7.89/1.33, pivot 3×10^{10}) is the version used across the recent literature; the in-repo `Greene20_Supple_5.csv` and `aa58_greene_supmat.pdf` confirmed all central values and uncertainties to the quoted precision. No fabricated numbers.

2. Methodology — “how many σ from the relation” is field standard (VERIFIED against other analyses)

For a single object the offset from a scaling relation is judged against the total dispersion = measurement uncertainty combined in quadrature with the relation’s intrinsic scatter (the population dispersion about the mean). An object is “consistent with the relation” if $|\Delta| \lesssim \sqrt{(\sigma_{\text{meas}}^2 + \epsilon_{\text{int}}^2)}$. Verified (2026-06-17, full text read in Zotero) against the closest analyses:

- Pacucci, Nguyen, Carniani, Maiolino & Fan 2023 (ApJL 957, L3) + Pacucci & Loeb 2024 (arXiv:2401.04159, read in full) — the reference high- z offset analysis. They quantify the offset of a point from the local relation and assess significance against the intrinsic scatter combined with the measurement errors: their “ $>3\sigma$ violation” of the $M_{\bullet}-M_{\star}$ relation holds “*unless... errors of a factor ~ 60 in their black hole mass or their stellar mass all in the same direction*” — i.e. exactly offset $\div$ (measurement $\oplus$ intrinsic scatter), the same construction we use. They fit the relation with a free intrinsic-scatteer term (Hogg, Bovy & Lang 2010

likelihood; orthogonal scatter ν) and correct for Malmquist-type bias (Lauer+2007) — the population-fit generalization of our single-point test.

- Melo-Carneiro et al. 2025 (Cosmic Horseshoe, MNRAS; the DIRECT analogue — a lensing+dynamics SMBH placed on $M_{\bullet}-\sigma$ at $z_l=0.44$, read in full) — 5σ BH detection; measurement uncertainty from 200 Monte Carlo realizations (standard deviation as the error); the BH is reported as an over-massive $M_{\bullet}-\sigma$ outlier with consistency quoted “within $N\sigma$ ” against the relation, same σ -framework.
- Reines & Volonteri 2015, Kormendy & Ho 2013 — report residuals about the relation relative to the (vertical) intrinsic scatter ϵ .

Convention check / nuances (honest): - Vertical vs orthogonal scatter. Pacucci fits the *relation itself* and uses *orthogonal* intrinsic scatter (because slope+scatter+points are inferred jointly). We instead place ONE point on a fixed, published relation, so the correct convention is the vertical offset $\Delta \log M_{\bullet}$ at fixed x compared to the relation’s vertical intrinsic scatter — which is exactly how Greene+2020 define ϵ (their form is $\log M_{\bullet} = \alpha + \beta \log x + \epsilon$, ϵ in the $\log M_{\bullet}$ direction). So vertical-offset $\div$ vertical- ϵ is internally consistent; mixing in Pacucci’s orthogonal ϵ would be the inconsistency, and we avoid it. - x -error propagation. Measurement error in the independent variable (σ_e , $M_{\star}$) is propagated vertically through the slope, $\beta \cdot \sigma_x$, then added in quadrature — standard. - Relation-parameter (α, β) errors. Optional; for a single object the measurement $\oplus$ intrinsic-scatter terms dominate. We report N_{σ} both without and with them ($0.57/0.69 \rightarrow 0.56/0.67$) — negligible. - Intrinsic scatter must be in the denominator for a single object. This is the crux and it is standard: ϵ is the population dispersion, so a lone galaxy is “on the relation” if it lies within $\sqrt{(\text{meas}^2 + \epsilon^2)}$. Reporting Δ/ϵ alone ($M_{\bullet}-\sigma$ 0.60, $M_{\bullet}-M_{\star}$ 0.75) gives essentially the same answer here because ϵ dominates.

Conclusion: our treatment is consistent with the field standard. We use the vertical single-point convention matched to Greene+2020’s published vertical ϵ (the appropriate choice when placing a point on a fixed relation), with measurement and x -errors propagated in quadrature — the same offset-vs-combined- scatter logic Pacucci+2023/2024 and Melo-Carneiro+2025 apply.

Our estimator (relation_offset_significance.py):

$$N_{\sigma} = |\Delta| / \sqrt{\sigma_{M_{\bullet}}^2 + (\beta \cdot \sigma_x)^2 + \epsilon_{\text{int}}^2 + [\sigma_{\alpha}^2 + (\sigma_{\beta} \cdot \log(x/\text{pivot}))^2]} \\ \Delta = \log M_{\bullet}(\text{obs}) - [\alpha + \beta \cdot \log(x_{\text{obs}}/\text{pivot})] \quad (\text{offset from the MEAN relation})$$

- $\sigma_{M_{\bullet}}$: our $\log M_{\text{BH}}$ measurement error (PROVISIONAL; Ferrami free-BH 1σ envelope)
- $\beta \cdot \sigma_x$: our $\sigma_e / \log M_{\star}$ error propagated through the slope
- ϵ_{int} : Greene+2020 intrinsic scatter (the dominant term)
- optional: the relation-parameter (α, β) uncertainties — the “full” variant

This is the pythagorean combination the analysis asked for. We report three numbers per relation: (i) the dex offset from the scatter-free mean line; (ii) the distance to that mean line in measurement- σ only; (iii) the consistency N_{σ} including intrinsic scatter (the field-standard single-object test).

3. Results (with $M_{\text{BH}} = 5.2_{+1.7/-1.4} \times 10^8$, PROVISIONAL; $\sigma_e=267$, $\log M_{\star}=11.46$)

relation	Δ (mean)	meas-only	N_{σ} (incl. ϵ_{int})	N_{σ} ($+\alpha, \beta$)	$\Delta/\epsilon_{\text{int}}$
$M_{\bullet}-\sigma$	-0.26 dex	1.75σ	0.57σ	0.56σ	0.60
$M_{\bullet}-M_{\star}$	-0.49 dex	1.79σ	0.69σ	0.67σ	0.75

($\Delta/\epsilon_{\text{int}}$ = offset in pure intrinsic-scatter units, the simplest field metric — agrees with the full quadrature because ϵ_{int} dominates.) Verdict: AGEL J0206 is a mildly under-massive BH relative to the *mean* early-type relations but lands well within their intrinsic scatter ($<0.7\sigma$) — i.e. consistent with both, as a genuine independent high- z ($z=0.676$) probe. The most distant calibrator in the Greene+2020 Early sample is A1836-BCG at 152 Mpc ($z \approx 0.034$), so AGEL J0206 extends the test $\sim 20\times$ in redshift.

4. Known inconsistency to reconcile (flagged, not yet changed)

The registry leaf `lens_model.mbh_sigma_offset` (-0.69) was computed against $[\text{mean} + \epsilon_{\text{int}}]$ — i.e. the intrinsic scatter was added to the mean relation before differencing. That is not the offset from the relation (the mean-line offset is -0.26), and it is not where the relation is drawn in the figure. Per user (2026-06-17) the figure scatter band is intentionally the mean-locus 1σ (how we choose to treat per-object scatter when plotting) and is kept; the registry offset reconciliation ($-0.69 \rightarrow -0.26$, add `mbh_mstar_offset` = -0.49 , add `n_sigma` leaves) is deferred pending authorization.

5. Redshift reach of the Greene+2020 calibrating sample (the independent-probe point)

The Greene+2020 relations are calibrated on local dynamical BH masses. The supplemental tables list distances (`Distance_Mpc`), not redshifts; converting with $z \approx H_0 \cdot D/c$ ($H_0 = 67.7$, Planck 2015) gives the redshift reach. The most distant calibrator in both relation samples is the same object:

relation	sample (as built in the figures)	N	most distant galaxy	distance	$z \approx H_0 D/c$
$M_\bullet - \sigma$	Suppl-2 (E+S0) + Suppl-3 (E) + Suppl-4 (S0)	83	A1836-BCG	152.4 Mpc	≈ 0.034
$M_\bullet - M_\star$	Greene Fig-10 E+S0 with $\log M_\star$	61	A1836-BCG	152.4 Mpc	≈ 0.034

Next most distant (both samples), all giant ellipticals/BCGs: NGC 6086 (138 Mpc, $z \approx 0.031$), NGC 7768 (116 Mpc, $z \approx 0.026$), NGC 4889 (102 Mpc, $z \approx 0.023$), NGC 3842 (92 Mpc, $z \approx 0.021$); the overwhelming majority sit far closer (Virgo/Coma and nearer). The full 120-galaxy Greene+2020 dynamical compilation also caps at A1836-BCG.

Source: `../AGEL_Mbh_sigma/Greene20_Supple_{2,3,4}.csv` (`Distance_Mpc` column), samples assembled exactly as in the paper figure code ($M_\bullet - \sigma$ = Suppl-2 HT $\in\{E, S0\}$ $\square$ Suppl-3 $\square$ Suppl-4 HT=S0; $M_\bullet - M_\star$ = Suppl-3 HT=E $\square$ Suppl-4 HT=S0 with finite $\log M_\star$). Reproduce: `python -m scripts.greene_sample_redshift`. Caveat: these are redshift-independent distances (SBF/Cepheid for the nearest, Hubble-flow for the farthest); the z conversion is exact only in the Hubble-flow limit — reliable for A1836-BCG (~ 3440 km/s) but a recession-velocity proxy for the nearest members where peculiar velocities dominate.

Why it matters: the local calibration extends only to $z \approx 0.034$. AGEL J0206 at $z = 0.67564$ is $\sim 20\times$ higher in redshift (and $\sim 4.4\times$ in comoving distance) than the most distant calibrator — the core justification for treating this as a genuinely independent, high- z probe of the relations.